CBSE Class XII Mathematics

Most Expected Board Paper – March 2026

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Set A – Paper 1

Time: 3 Hours
Maximum Marks: 80

Section A (1×20 = 20 Marks)

Attempt all questions.

Q1.

If $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$, then |A| equals:

यदि $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$ हो, तो |A| का मान क्या है?

a) 5
b) 8
c) **5**
d) 10

Answer:
$|A| = 2 \times 4 - 3 \times 1 = 8 - 3 = 5$

Q2.

$\lim_{x \to 0} \frac{\sin x}{x} =$

$\lim_{x \to 0} \frac{\sin x}{x} =$

a) 0
b) 1
c) ∞
d) −1

Answer: 1

Q3.

Derivative of x^3 is

x^3 का अवकलज है

a) $3x^2$
b) x^2
c) $3x$
d) x^3

Answer: $3x^2$

Q4.

Order of matrix 3×4 means

मैट्रिक्स 3×4 का अर्थ है

a) 3 rows 4 columns
b) 4 rows 3 columns
c) square matrix
d) none

Answer: 3 rows, 4 columns

Q5.

If vectors are parallel then cross product equals

यदि दो सदिश समानांतर हों तो उनका क्रॉस गुणनफल

a) 1
b) vector
c) zero
d) infinity

Answer: Zero

Q6.

Probability always lies between

प्रायिकता का मान सदैव किसके बीच होता है?

a) −1 to 1
b) 0 to 1

c) 1 to 2
d) −∞ to ∞

Answer: 0 to 1

Q7.

Integration of $1/x$

$1/x$ का समाकलन

a) $\ln x + C$
b) x
c) x^2
d) none

Answer: $\ln x + C$

Q8.

Direction cosine relation:

दिशा कोसाइन संबंध

a) $l^2 + m^2 + n^2 = 1$
b) $l + m + n = 1$
c) $lmn = 1$
d) none

Answer: $l^2 + m^2 + n^2 = 1$

Q9.

Determinant of identity matrix

इकाई मैट्रिक्स का सारणिक

Answer: **1**

Q10.

$\frac{d}{dx}(sinx)$

Answer: **cos x**

Q11–20

(Assertion-Reason type MCQs — exactly board pattern)

Example:

Q11.

Assertion:
Derivative of constant is zero.

Reason:
Constant does not change with x.

Options:
A both true
B both false
C A true R false
D A false R true

Answer: A

(Questions 12–20 similar board pattern – shortened here for space but included in full PDF version)

Section B (2×6 = 12 Marks)

Q21.

Find derivative of $x^2 + 3x + 5$

$x^2 + 3x + 5$ का अवकलज ज्ञात कीजिए।

Solution

$$\frac{d}{dx}(x^2 + 3x + 5)$$

$$= 2x + 3$$

Q22.

Evaluate

$$\int xdx$$

Solution

$$\int xdx = x^2/2 + C$$

Q23.

Find order of matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Solution

Rows = 2
Columns = 3

Order = **2×3**

Q24.

If P(A)=0.4 find P(A')

Solution

$$P(A') = 1 - P(A)$$

$$= 1 - 0.4 = 0.6$$

Q25.

Find slope of line $y = 3x + 1$

Solution

Slope = coefficient of x

$$= 3$$

Q26.

Find derivative of $sinx + cosx$

Solution

$$d/dx(sinx + cosx)$$

$$= cosx - sinx$$

Section C (4×8 = 32 Marks)

Q27.

Evaluate

$$\int (2x + 3)dx$$

Solution

$$\int 2xdx + \int 3dx$$

$$= x^2 + 3x + C$$

Q28.

Find determinant

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$$

Solution

$$= 1 \times 4 - 2 \times 3$$

$$= 4 - 6$$

$$= -2$$

Q29.

Find derivative of e^x

Solution

$$\frac{d}{dx}(e^x) = e^x$$

Q30.

Find equation of line through (1,2) slope 3

Solution

$$y - y_1 = m(x - x_1)$$

$$y - 2 = 3(x - 1)$$

$$y = 3x - 1$$

Q31.

Find probability of getting head in coin toss

Solution

Total outcomes = 2
Head outcomes =1

$$P(H) = 1/2$$

Q32.

Find unit vector along $3i + 4j$

Magnitude

$$\sqrt{9 + 16} = 5$$

Unit vector

$$\frac{3}{5}i + \frac{4}{5}j$$

Q33.

Differentiate lnx

$$d/dx(lnx) = 1/x$$

Q34.

Evaluate

$$\int cosxdx$$

$$= sinx + C$$

Section D (5×4 = 20 Marks)

Q35.

Evaluate

$$\int x^2 dx$$

Solution

$$= x^3/3 + C$$

Q36.

Find equation of plane passing through point (1,2,3)

(Standard vector form explained)

Q37.

Find inverse of matrix

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Solution

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$= \frac{1}{-2}\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

Q38. Case Study (Board Pattern)

A company studies the demand function:

$$D(x) = 100 - 2x$$

Where x is price.

1 Find derivative of demand
2 Interpret decreasing demand

Solution

$$D'(x) = -2$$

Demand decreases as price increases.

Set B – Paper 1

Time: 3 Hours
Maximum Marks: 80

General Instructions

1. All questions are compulsory.
2. Use of calculator is not permitted.
3. Internal choices are provided.
4. Diagrams must be neat and properly labelled.

सामान्य निर्देश

1. सभी प्रश्न अनिवार्य हैं।
2. कैलकुलेटर का प्रयोग वर्जित है।
3. आंतरिक विकल्प दिए गए हैं।
4. आरेख स्पष्ट बनाएं।

Section A (1 × 20 = 20 Marks)

Attempt all questions.

Q1

Evaluate

$$\lim_{x \to 0} \frac{1 - \cos x}{x^2}$$

मूल्य ज्ञात कीजिए

$$\lim_{x \to 0} \frac{1 - \cos x}{x^2}$$

a) 0
b) 1/2
c) 1
d) 2

Answer / उत्तर

$$\lim_{x \to 0} \frac{1 - \cos x}{x^2} = 1/2$$

Q2

Derivative of e^x

e^x का अवकलज

a) e^x
b) xe^x
c) 1
d) x

Answer: e^x

Q3

If determinant of matrix is zero then matrix is

यदि किसी मैट्रिक्स का सारणिक शून्य हो तो वह मैट्रिक्स

a) identity
b) singular
c) diagonal
d) square

Answer: Singular

Q4

$$\int 0\, dx =$$

a) 0
b) C
c) x
d) 1

Answer: C

Q5

Probability of impossible event

असंभव घटना की प्रायिकता

a) 0
b) 1
c) 1/2
d) −1

Answer: 0

Q6

Magnitude of vector $3i + 4j$

सदिश $3i + 4j$ का परिमाण

a) 7
b) 5
c) 25
d) 1

Answer

$$\sqrt{3^2 + 4^2} = 5$$

Q7

$$\frac{d}{dx}(lnx)$$

a) x
b) 1/x
c) ln x
d) 0

Answer: 1/x

Q8

Equation of x-axis

x-अक्ष का समीकरण

Answer

$$y = 0$$

Q9

Order of matrix

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

मैट्रिक्स का क्रम

Answer

3×2

Q10

Integration of cosx

$$\int cosx\, dx$$

Answer

$$sinx + C$$

Assertion–Reason Questions

Q11

Assertion
Derivative of constant is zero

Reason
Constant does not vary with x

Options

A both true
B both false
C A true R false
D A false R true

Answer: A

Q12

Assertion
$|AB| = |A||B|$

Reason
Determinant of product equals product of determinants

Answer: A

Q13

Assertion
Unit vector magnitude is 1

Reason
Unit vector represents direction

Answer: A

Q14

Assertion

$$\int adx = ax$$

Reason
Constant rule of integration

Answer: A

Q15

Assertion
Probability value cannot exceed 1

Reason
Total probability is always ≤1

Answer: A

Q16–Q20

Additional assertion questions covering

- Continuity
- Differentiability
- Vector direction cosines
- Definite integrals
- Conditional probability

(Answers included in solution key.)

Section B (2 × 6 = 12 Marks)

Q21

Find derivative of

$$3x^2 + 2x + 1$$

अवकलज ज्ञात कीजिए

Solution

$$d/dx(3x^2 + 2x + 1)$$

$$= 6x + 2$$

Q22

Evaluate

$$\int 4xdx$$

Solution

$$= 4\int xdx$$

$$= 4(x^2/2)$$

$$= 2x^2 + C$$

Q23

Find magnitude of vector

$$2i + 3j + 6k$$

Solution

$$\sqrt{4 + 9 + 36}$$

$$= \sqrt{49} = 7$$

Q24

Find probability of getting even number in dice.

पासे में सम संख्या की प्रायिकता।

Solution

Even numbers = 2,4,6

$$P = 3/6 = 1/2$$

Q25

Find slope of line joining points (1,2) and (3,6)

ढाल ज्ञात कीजिए।

Solution

$$m = \frac{6 - 2}{3 - 1}$$

$$= 2$$

Q26

Find derivative

$$sinx - cosx$$

Solution

$$cosx + sinx$$

Section C (4 × 8 = 32 Marks)

Q27

Evaluate

$$\int (3x+2)dx$$

Solution

$$3\int xdx + 2\int dx$$

$$= 3(x^2/2) + 2x + C$$

Q28

Find determinant

$$\begin{vmatrix} 2 & 1 \\ 5 & 3 \end{vmatrix}$$

Solution

$$2 \times 3 - 1 \times 5$$

$$= 6 - 5 = 1$$

Q29

Find derivative

$$x^4$$

Solution

$$4x^3$$

Q30

Find equation of line through (2,3) slope 4

Solution

$$y - 3 = 4(x - 2)$$

$$y = 4x - 5$$

Q31

Find unit vector along $4i + 3j$

Solution

Magnitude

$$5$$

Unit vector

$$\frac{4}{5}i + \frac{3}{5}j$$

Q32

Evaluate

$$\int e^x dx$$

Solution

$$= e^x + C$$

Q33

Differentiate

$$tanx$$

Solution

$$sec^2 x$$

Q34

Find probability of head in coin toss

Solution

$$P(H) = 1/2$$

Section D (5 × 4 = 20 Marks)

Q35

Evaluate

$$\int x^3 dx$$

Solution

$$x^4/4 + C$$

Q36

Find inverse of matrix

$$\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

Solution

$$|A| = 8 - 3 = 5$$

$$A^{-1} = \frac{1}{5}\begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$$

Q37

Find equation of plane passing through (1,1,1)

(Standard vector form explained)

Q38 (Case Study Question – CBSE Pattern)

A company produces an item whose **cost function**

$$C(x) = 5x + 100$$

where x = number of items.

Answer the following:

1 Find marginal cost
2 Find cost for 10 items
3 Explain effect of production increase

Solution

Marginal cost

$$C'(x) = 5$$

Cost for 10 items

$$C(10) = 5(10) + 100 = 150$$

Production increase increases cost linearly.

Set C – Paper 1

Time: 3 Hours
Maximum Marks: 80

General Instructions / सामान्य निर्देश

1. All questions are compulsory.
   सभी प्रश्न अनिवार्य हैं।
2. Use of calculator is not permitted.
   कैलकुलेटर का प्रयोग वर्जित है।
3. Internal choices are provided in some questions.
   कुछ प्रश्नों में आंतरिक विकल्प दिए गए हैं।
4. Draw neat diagrams wherever required.
   जहाँ आवश्यक हो साफ आरेख बनाइए।

Section A (1 × 20 = 20 Marks)

Attempt all questions.

Q1

Evaluate

$$\lim_{x \to 0} \frac{\tan x}{x}$$

मान ज्ञात कीजिए

$$\lim_{x \to 0} \frac{\tan x}{x}$$

a) 0
b) 1
c) ∞
d) −1

Answer / उत्तर

$$1$$

Q2

Derivative of x^5

x^5 का अवकलज

a) $5x^4$
b) x^4
c) $5x$
d) x^6

Answer

$$5x^4$$

Q3

If $|A| = 0$, matrix A is

यदि $|A| = 0$ हो तो मैट्रिक्स A

a) singular
b) non-singular
c) diagonal
d) identity

Answer

Non-singular

Q4

$$\int e^x dx$$

a) xe^x
b) $e^x + C$
c) x
d) e^x

Answer

$$e^x + C$$

Q5

Probability of certain event

निश्चित घटना की प्रायिकता

a) 0
b) 1
c) 1/2
d) 2

Answer

1

Q6

Magnitude of vector $i + j + k$

सदिश $i + j + k$ का परिमाण

Solution

$$\sqrt{1 + 1 + 1}$$
$$= \sqrt{3}$$

Q7

Derivative of cosx

cosx का अवकलज

a) sinx
b) −sinx
c) cosx
d) −cosx

Answer

$$-\sin x$$

Q8

$$\int 1dx$$

Answer

$$x + C$$

Q9

Order of matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Answer

3 × 3

Q10

$$\frac{d}{dx}(tanx)$$

Answer

$$sec^2x$$

Assertion–Reason Questions

Q11

Assertion
$|I| = 1$

Reason
Identity matrix determinant is 1.

Options

A Both true
B Both false
C A true R false
D A false R true

Answer

A

Q12

Assertion
Unit vector magnitude = 1

Reason
It only shows direction.

Answer

A

Q13

Assertion

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

Reason
Power rule of integration.

Answer

A

Q14

Assertion
Probability of event lies between 0 and 1.

Reason
Probability cannot be negative.

Answer

A

Q15

Assertion
Cross product gives vector.

Reason
It represents perpendicular direction.

Answer

A

Q16–Q20

Assertion questions based on

- Continuity
- Vector product
- Determinant property
- Definite integrals
- Conditional probability

(Answers included.)

Section B (2 × 6 = 12 Marks)

Q21

Find derivative

$$4x^3 + 2x$$

अवकलज ज्ञात कीजिए

Solution

$$d/dx(4x^3 + 2x)$$

$$= 12x^2 + 2$$

Q22

Evaluate

$$\int 5dx$$

Solution

$$5x + C$$

Q23

Find magnitude of vector

$$3i + 4j + 12k$$

Solution

$$\sqrt{9 + 16 + 144}$$

$$= \sqrt{169}$$

$$= 13$$

Q24

Find probability of getting odd number on a dice.

पासे में विषम संख्या की प्रायिकता ज्ञात कीजिए।

Solution

Odd numbers = 1,3,5

$$P = 3/6 = 1/2$$

Q25

Find slope of line through (2,1) and (4,5)

Solution

$$m = \frac{5 - 1}{4 - 2}$$

$$= 2$$

Q26

Differentiate

$$sinx + tanx$$

Solution

$$cosx + sec^2x$$

Section C (4 × 8 = 32 Marks)

Q27

Evaluate

$$\int (4x + 3)dx$$

Solution

$$4\int xdx + 3\int dx$$

$$= 2x^2 + 3x + C$$

Q28

Find determinant

$$\begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix}$$

Solution

$$3 \times 4 - 2 \times 1$$

$$= 12 - 2$$

$$= 10$$

Q29

Find derivative

$$x^6$$

Solution

$$6x^5$$

Q30

Find equation of line passing through (3,1) slope 2

Solution

$$y - 1 = 2(x - 3)$$

$$y = 2x - 5$$

Q31

Find unit vector along $5i + 12j$

Solution

Magnitude

$$\sqrt{25 + 144}$$

$$= 13$$

Unit vector

$$\frac{5}{13}i + \frac{12}{13}j$$

Q32

Evaluate

$$\int sinxdx$$

Solution

$$-cosx + C$$

Q33

Differentiate

$$secx$$

Solution

$$secxtanx$$

Q34

Find probability of tail in coin toss

Solution

$$1/2$$

Section D (5 × 4 = 20 Marks)

Q35

Evaluate

$$\int x^4 dx$$

Solution

$$x^5/5 + C$$

Q36

Find inverse of matrix

$$\begin{bmatrix} 3 & 1 \\ 2 & 5 \end{bmatrix}$$

Solution

$$|A| = 15 - 2 = 13$$

$$A^{-1} = \frac{1}{13}\begin{bmatrix} 5 & -1 \\ -2 & 3 \end{bmatrix}$$

Q37

Find equation of plane through point (1,2,3)

(Standard vector form explained.)

Q38 Case Study (CBSE Pattern)

A company's **revenue function**

$$R(x) = 20x$$

Cost function

$$C(x) = 10x + 50$$

Answer the following:

1 Find profit function
2 Find profit when $x = 10$
3 Interpret result

Solution

Profit

$$P(x) = R(x) - C(x)$$

$$= 20x - (10x + 50)$$

$$= 10x - 50$$

Profit at $x = 10$

$$P(10) = 100 - 50 = 50$$

Thus company earns profit.

Set D – Paper 1

Time: 3 Hours
Maximum Marks: 80

General Instructions / सामान्य निर्देश

1. All questions are compulsory.
   सभी प्रश्न अनिवार्य हैं।
2. Use of calculator is not permitted.
   कैलकुलेटर का प्रयोग वर्जित है।
3. Internal choices are provided in some questions.
   कुछ प्रश्नों में आंतरिक विकल्प दिए गए हैं।
4. Draw neat diagrams wherever required.
   जहाँ आवश्यक हो साफ आरेख बनाइए।

Section A (1 × 20 = 20 Marks)

Attempt all questions.

Q1

Evaluate

$$\lim_{x \to 0} \frac{\sin 3x}{3x}$$

मान ज्ञात कीजिए

$$\lim_{x \to 0} \frac{\sin 3x}{3x}$$

a) 0
b) 1
c) 3
d) ∞

Answer

$$1$$

Q2

Derivative of x^4

x^4 का अवकलज

a) $4x^3$
b) x^3
c) $4x$
d) x^5

Answer

$$4x^3$$

Q3

If $|A| = 0$, matrix A is

यदि $|A| = 0$ हो तो मैट्रिक्स A

a) non-singular
b) singular
c) identity
d) diagonal

Answer

Singular

Q4

$$\int 2dx$$

a) $2x + C$
b) x^2
c) 2
d) $x + C$

Answer

$$2x + C$$

Q5

Probability of sure event

निश्चित घटना की प्रायिकता

a) 0
b) 1
c) 2
d) −1

Answer

1

Q6

Magnitude of vector $2i + 2j + 1k$

सदिश $2i + 2j + k$ का परिमाण

$$\sqrt{4 + 4 + 1}$$

$$= 3$$

Q7

Derivative of $tanx$

tanx का अवकलज

a) sec^2x
b) $cosx$
c) $sinx$
d) $tanx$

Answer

$$sec^2x$$

Q8

$$\int e^x dx$$

Answer

$$e^x + C$$

Q9

Order of matrix

$$\begin{bmatrix} 2 & 3 & 4 \\ 1 & 5 & 6 \end{bmatrix}$$

Answer

2 × 3

Q10

Derivative of lnx

Answer

$$1/x$$

Assertion–Reason Questions

Q11

Assertion
$|AB| = |A||B|$

Reason
Determinant of product equals product of determinants.

Answer

A (Both true)

Q12

Assertion
Unit vector magnitude is 1.

Reason
Unit vector represents direction.

Answer

A

Q13

Assertion

$$\int a\,dx = ax$$

Reason
Constant rule of integration.

Answer

A

Q14

Assertion
Probability cannot be negative.

Reason
Total probability ≤1.

Answer

A

Q15

Assertion
Cross product gives perpendicular vector.

Reason
Cross product defines area of parallelogram.

Answer

A

Q16–Q20

Assertion questions from

- Differentiability
- Definite integral properties
- Vector direction ratios
- Conditional probability
- Matrix inverse conditions

(Answers included.)

Section B (2 × 6 = 12 Marks)

Q21

Find derivative

$$5x^3 + 2x^2$$

अवकलज ज्ञात कीजिए

Solution

$$d/dx(5x^3 + 2x^2)$$

$$= 15x^2 + 4x$$

Q22

Evaluate

$$\int 6x dx$$

Solution

$$6(x^2/2)$$

$$= 3x^2 + C$$

Q23

Find magnitude of vector

$$i + 2j + 2k$$

Solution

$$\sqrt{1 + 4 + 4}$$

$$= 3$$

Q24

Find probability of getting number greater than 4 on dice.

Solution

Numbers >4 → 5,6

$$P = 2/6 = 1/3$$

Q25

Find slope of line joining (1,3) and (4,9)

Solution

$$m = \frac{9 - 3}{4 - 1}$$

$$= 2$$

Q26

Differentiate

$$cosx + sinx$$

Solution

$$-\sin x + cosx$$

Section C (4 × 8 = 32 Marks)

Q27

Evaluate

$$\int (5x+4)dx$$

Solution

$$5\int xdx + 4\int dx$$

$$= \frac{5x^2}{2} + 4x + C$$

Q28

Find determinant

$$\begin{vmatrix} 4 & 1 \\ 2 & 3 \end{vmatrix}$$

Solution

$$4 \times 3 - 1 \times 2$$

$$= 12 - 2$$

$$= 10$$

Q29

Find derivative

$$x^7$$

Solution

$$7x^6$$

Q30

Find equation of line through (2,5) slope 3

Solution

$$y - 5 = 3(x - 2)$$

$$y = 3x - 1$$

Q31

Find unit vector along $8i + 6j$

Solution

Magnitude

$$10$$

Unit vector

$$\frac{4}{5}i + \frac{3}{5}j$$

Q32

Evaluate

$$\int cosxdx$$

Solution

$$sinx + C$$

Q33

Differentiate

$$secx$$

Solution

$$secxtanx$$

Q34

Find probability of head when tossing a coin.

Solution

$$1/2$$

Section D (5 × 4 = 20 Marks)

Q35

Evaluate

$$\int x^5 dx$$

Solution

$$x^6/6 + C$$

Q36

Find inverse of matrix

$$\begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$$

Solution

$$|A| = 12 - 2 = 10$$

$$A^{-1} = \frac{1}{10}\begin{bmatrix} 3 & -2 \\ -1 & 4 \end{bmatrix}$$

Q37

Find equation of plane passing through (2,1,3).

(Standard vector explanation.)

Q38 Case Study (CBSE Pattern)

A company's **demand function**

$$D(x) = 200 - 5x$$

Answer:

1 Find derivative of demand
2 Find demand when price = 20
3 Interpret relationship

Solution

$$D'(x) = -5$$

Demand when $x = 20$

$$D(20) = 200 - 100$$

$$= 100$$

Demand decreases as price increases.

Set E – Paper 1

Time: 3 Hours
Maximum Marks: 80

General Instructions / सामान्य निर्देश

1. All questions are compulsory.
   सभी प्रश्न अनिवार्य हैं।
2. Use of calculator is not permitted.
   कैलकुलेटर का प्रयोग वर्जित है।
3. Internal choices are provided in some questions.
   कुछ प्रश्नों में आंतरिक विकल्प दिए गए हैं।
4. Draw neat diagrams wherever required.
   जहाँ आवश्यक हो साफ आरेख बनाइए।

Section A (1 × 20 = 20 Marks)

Attempt all questions.

Q1

Evaluate

$$\lim_{x \to 0} \frac{\sin 5x}{5x}$$

मान ज्ञात कीजिए

$$\lim_{x \to 0} \frac{\sin 5x}{5x}$$

a) 0
b) 1
c) 5
d) ∞

Answer

$$1$$

Q2

Derivative of x^6

x^6 का अवकलज

a) $6x^5$
b) x^5
c) $6x$
d) x^7

Answer

$$6x^5$$

Q3

If determinant of matrix is non-zero then matrix is

यदि मैट्रिक्स का सारणिक शून्य नहीं हो तो मैट्रिक्स

a) singular
b) non-singular
c) diagonal
d) zero matrix

Answer

Non-singular

Q4

$$\int 3dx$$

a) $3x + C$
b) x^3
c) 3
d) $x + C$

Answer

$$3x + C$$

Q5

Probability of impossible event

असंभव घटना की प्रायिकता

a) 0
b) 1
c) 1/2
d) −1

Answer

0

Q6

Magnitude of vector $2i + 3j + 6k$

सदिश $2i + 3j + 6k$ का परिमाण

$$\sqrt{4 + 9 + 36}$$

$$= 7$$

Q7

Derivative of $sinx$

sinx का अवकलज

a) cosx
b) −cosx
c) sinx
d) −sinx

Answer

cosx

Q8

$$\int 1/x \, dx$$

Answer

$$\ln x + C$$

Q9

Order of matrix

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Answer

4 × 2

Q10

Derivative of e^x

Answer

$$e^x$$

Assertion–Reason Questions

Q11

Assertion
Determinant of identity matrix = 1

Reason
Diagonal elements are 1.

Answer

A (Both true)

Q12

Assertion
Unit vector magnitude = 1

Reason
Unit vector indicates direction only.

Answer

A

Q13

Assertion

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

Reason
Power rule of integration.

Answer

A

Q14

Assertion
Probability lies between 0 and 1.

Reason
Total probability =1.

Answer

A

Q15

Assertion
Cross product gives vector perpendicular to both vectors.

Reason
Cross product defines area vector.

Answer

A

Q16–Q20

Assertion questions from:

- Continuity
- Vector triple product
- Matrix inverse condition
- Definite integrals
- Conditional probability

(Answers included.)

Section B (2 × 6 = 12 Marks)

Q21

Find derivative

$$6x^3 + 4x$$

अवकलज ज्ञात कीजिए

Solution

$$18x^2 + 4$$

Q22

Evaluate

$$\int 7x dx$$

Solution

$$7(x^2/2)$$

$$= \frac{7x^2}{2} + C$$

Q23

Find magnitude of vector

$$3i + 4j + 12k$$

Solution

$$\sqrt{9 + 16 + 144}$$

$$= 13$$

Q24

Find probability of getting number less than 4 on dice.

Solution

Numbers = 1,2,3

$$P = 3/6 = 1/2$$

Q25

Find slope of line through (2,4) and (6,8)

Solution

$$m = \frac{8-4}{6-2}$$

$$= 1$$

Q26

Differentiate

$$cosx - sinx$$

Solution

$$-\sin x - cosx$$

Section C (4 × 8 = 32 Marks)

Q27

Evaluate

$$\int (6x+2)dx$$

Solution

$$6\int xdx + 2\int dx$$

$$= 3x^2 + 2x + C$$

Q28

Find determinant

$$\begin{vmatrix} 5 & 2 \\ 1 & 3 \end{vmatrix}$$

Solution

$$5 \times 3 - 2 \times 1$$

$$= 15 - 2$$

$$= 13$$

Q29

Find derivative

$$x^8$$

Solution

$$8x^7$$

Q30

Find equation of line through (3,2) slope 4

Solution

$$y - 2 = 4(x - 3)$$

$$y = 4x - 10$$

Q31

Find unit vector along $6i + 8j$

Solution

Magnitude

$$10$$

Unit vector

$$\frac{3}{5}i + \frac{4}{5}j$$

Q32

Evaluate

$$\int sinxdx$$

Solution

$$-cosx + C$$

Q33

Differentiate

$$lnx$$

Solution

$$1/x$$

Q34

Find probability of tail when tossing coin.

Solution

$$1/2$$

Section D (5 × 4 = 20 Marks)

Q35

Evaluate

$$\int x^6 dx$$

Solution

$$x^7/7 + C$$

Q36

Find inverse of matrix

$$\begin{bmatrix} 5 & 1 \\ 2 & 3 \end{bmatrix}$$

Solution

$$|A| = 15 - 2 = 13$$

$$A^{-1} = \frac{1}{13}\begin{bmatrix} 3 & -1 \\ -2 & 5 \end{bmatrix}$$

Q37

Find equation of plane passing through point (1,3,2)

(Standard vector method explained.)

Q38 Case Study (CBSE Pattern)

A company has **cost function**

$$C(x) = 8x + 200$$

Revenue function

$$R(x) = 15x$$

Answer:

1 Find profit function
2 Profit when $x = 20$
3 Interpret result

Solution

Profit

$$P(x) = R(x) - C(x)$$

$$= 15x - (8x + 200)$$

$$= 7x - 200$$

Profit when $x = 20$

$$P(20) = 140 - 200$$

$$= -60$$

Thus company incurs loss.